Spring Boot Logging

Dependencies

- `spring-boot-starter` **includes** **`spring-boot-starter-logging` by default**
- Uses **Logback + SLF4J** under the hood
- Features:
  - Formatted logs
  - Concurrent file access
  - Multiple destinations
  - Configurable log levels **without changing code**

Log Levels

- Control **verbosity** → helps with **debugging** and **monitoring**

Level	Purpose
TRACE	Most detailed, granular debugging
DEBUG	Development/debugging information
INFO	General runtime/system events
WARN	Potential issues
ERROR	Errors requiring attention
FATAL	Critical errors, may stop system

Example usage:

```
log.info("Application started");
log.warn("Potential issue detected");
log.error("An error occurred!");
```

Configuring Log Levels

1️⃣ Root Level (Default: INFO)

```
logging:
  level:
    root: WARN
```

2️⃣ Package-Specific Levels

```
logging:
  level:
    root: WARN
    org.springframework: INFO
    com.nbicocchi: DEBUG
```

- Use **higher levels** (WARN/ERROR) to reduce logging overhead
- Temporarily lower levels (DEBUG/TRACE) for **more verbose logs** in production

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Other Configurations

Log to a File

```
logging:
  file:
    name: app.log
```

Change Log Date Format

```
logging:
  pattern:
    dateformat: yyyy-MM-dd
```

Key Takeaways

- Spring Boot **simplifies logging** with built-in starter
- **SLF4J + Logback** provides a flexible logging API
- **Log levels** and **configuration via YAML** allow fine-grained control
- **Production vs Development** → adjust verbosity with minimal code changes

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Resources

- Spring Boot Logging
- SLF4J Introduction
- Java Logging Introduction